

Detection of citrus black spot disease and ripeness level in orange fruit using learning-to-augment incorporated deep networks

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ABSTRACT

Fruit infected by pests or diseases and fruit harvests with different levels of ripeness cause a lack of marketability, decrease in economic value, and increase in crop waste. In this study, we propose a robust and generalized deep convolutional neural network (CNN) model via fine-tuning the pre-trained models for detecting black spot disease and ripeness levels in orange fruit. A dataset containing 1896 confirmed orange images in the farm in four classes (unripe, half-ripe, ripe, and infected with black spot disease) was used. In order to prevent overfitting and increase the robustness and generalizability of the model, instead of using fundamental data augmentation techniques, a novel learning-to-augment strategy that creates new data using noisy and restored images was employed. Controllers using the Bayesian optimization algorithm were utilized to select the optimal noise parameters of Gaussian, speckle, Poisson, and salt-and-pepper noise to generate new noisy images. A convolutional autoencoder model was developed to produce newly restored images affected by optimized noise density. The dataset augmented by the best policies of the learning-to-augment strategy was used to fine-tune several pre-trained models (GoogleNet, ResNet18, ResNet50, ShuffleNet, MobileNetv2, and DenseNet201). The results showed that the learning-to-augment strategy for the fine-tuned ResNet50 achieved the best performance with 99.5% accuracy, and 100% F-measure by assigning images infected with black spot disease as the positive class. The proposed automatic disease and fruit quality monitoring technique can be also used for the detection of other diseases in agriculture and forestry.

1. Introduction

The existence of various diseases and pests in fruit and harvesting fruit at different levels of ripeness, are detrimental factors that lower the marketability of fruit, reduce their economic value, and increase fruit waste. Customers want high-quality fruit in terms of ripeness and select the ones with no pest damage. In the fruit and food industries, it is very important to carefully examine and analyze fruit's quality and other parameters (in terms of ripeness, diseases, etc.) and then deliver a high-quality product to the customer. Diagnosis of pests and different levels of fruit ripeness at the harvest stage is usually made by human experts, but sometimes experts may not be able to accurately distinguish fruit quality

for various reasons such as fatigue or lack of knowledge. Therefore, intelligent techniques such as image processing and artificial intelligence can be used to improve fruit evaluation and grading at harvest, considering their level of ripeness and pest/disease damage conditions (Azarmdel et al., 2020).

One of the many challenges that citrus growers face is the spread of citrus black spot disease that threaten fruit marketability and tree health. Citrus black spot symptoms can vary in appearance but all result in cosmetic lesions on the fruit rind. The most typical lesion, known as hard spot, starts as small brick red spots with black margins that increase in size and develop necrotic tissue in their center. These lesions render the fruit unacceptable for many important fresh markets. In severe cases,

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Fig. 1. Example of orange images in four classes: A) unripe, B) half-ripe, C) ripe, and D) infected with black spot disease.

Table 1

The specifications of the camera used to capture the orange fruit images.

Property	Value	Property	Value
Camera maker	Samsung	Dimensions	4128 × 2322
Camera model	SM-J500H	Horizontal resolution	72 dpi
F-stop	1.9	Vertical resolution	72 dpi
Exposure time	1/100 s.	Bit depth	24
ISO speed	ISO-64	Resolution unit	2
Exposure bias	0 step	Color representation	sRGB
Focal length	4 mm	Brightness	5.14
Max aperture	1.85	Exposure program	Normal
Metering mode	Center Weighted Average	White balance	Auto
Flash mode	No flash	EXIF version	220
35 mm focal length	28		

Table 2

Different noise definitions.

Notation	Definition
μ	Mean (speckle noise and Gaussian noise)
σ	Variance (speckle noise and Gaussian noise)
d	Noise density (salt-and-pepper)

citrus black spot diseases may cause premature fruit drop that reduces fruit yield. This disease must be effectively managed in the field to minimize its occurrence and further spread. In addition, fresh fruit packers and shippers must be careful not to ship any diseased fruit to regions or countries that prohibit it. Such a violation would harm future access to those markets (Bulanon et al., 2013; Xie and Lee, 2021).

In today's world of rapid work, it is essential to carry out different

industrial activities as quickly as possible, and thus developing automated systems in the fruit and food industry seems very important and efficient (Barbosa et al., 2021; Momeny et al., 2020). There are several methods for harvesting fruit; picking fruit can be done by hand or by special machines. Timely harvesting based on product quality (such as level of ripeness and fruit quality conditions) is critical and plays a significant role in controlling waste and marketing the products. In several studies, researchers have reported the texture and color of the fruit as indicators of its maturity and pest-freeness (Zhou et al., 2021). Through image processing and machine vision, new knowledge has been achieved that helps control the quality of food and agricultural products with high accuracy (Ji and Wu, 2022; Khosravi et al., 2021).

Researchers and developers are now equipped with machine learning (ML) to solve real-world problems computationally (Momeny et al., 2022). ML enables computers to build algorithms, recognize patterns in data, make predictions based on the data, and take action without being explicitly programmed. Different fields benefit from ML-based systems, including agriculture, food industry, etc. (He et al., 2022; Jahanbakhshi and Kheiralipour, 2020; Yang et al., 2019). An ML-based method that is most frequently applied today is deep learning (DL). Two essential properties of DL are its high abstraction levels and ability to learn image patterns automatically. A CNN is a type of DL architecture beneficial for image processing. CNNs are artificial neural networks (ANNs) that apply convolution operations in at least one of their layers (Jahanbakhshi et al., 2021c; Naranjo-Torres et al., 2020).

Much research has been recently conducted on the topic of CNN in object detection and image classification (Costa et al., 2021; Jahanbakhshi et al., 2021a; Momeny et al., 2021a; Partel et al., 2019). CNN is used because it can extract features from an input image. Compared with traditional classification algorithms, CNN can directly feed the image into the network to avoid pre-processing as well as the feature extraction process (Jahanbakhshi et al., 2021b). The three main layers of classical CNN are the convolutional layers, pooling, and fully connected (FC) layers. After winning the ImageNet competition, CNN received much

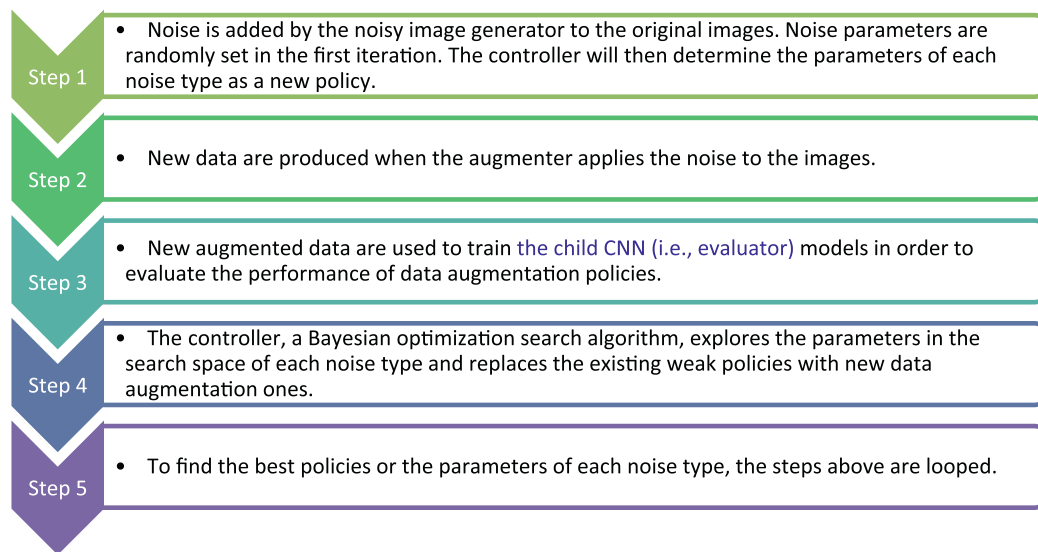


Fig. 2. The pipeline of the noisy-image-based learning-to-augment strategy.

attention. Soon, different CNN models were developed by several scientists, with various depth and width sizes for layers. An instance of layer replacement is when a conventional layer replaces a pooling layer with a higher number of strides in order to simplify the CNN architecture or reduce dimensions (Jahanbakhshi et al., 2020; Sharif et al., 2019).

Ashtiani et al. used DL models to identify ripeness stages for berries. Their study used CNN models including DenseNet, AlexNet, Inception-v3, ResNet-18, and ResNet-50. They also used transitional learning to fine-tune models and increase classification performance. They reported that the accuracy of AlexNet and ResNet-18 networks with 98.32% and 98.65% respectively showed the best performance for classifying ripeness levels in white berries and blackberries (Ashtiani et al., 2021). Using ANN and support vector machines (SVM), Azarmdel et al. tried to classify berry fruit according to its ripeness level (they identified three ripeness levels: unripe, ripe, and overripe) in order to control waste and improve the marketability of the fruit. Their research showed that compared to the SVM classification method, the ANN classification method was able to classify berry fruit with 100% accuracy (Azarmdel et al., 2020). Khojastehnazhand et al. diagnosed the maturity and estimated the volume of apricots using an image processing technique. They reported a statistically significant difference between ripeness stages according to the features of G, grayscale, L*, and b*. Also, LDA and QDA classifiers could classify apricot fruit with 90.4% and 92.3% accuracy based on color characteristics (Khojastehnazhand et al., 2019). Wan et al. proposed a method for detecting ripeness levels in tomatoes using computer vision. Their analysis of the results showed that the average detection accuracy for three maturity levels of tomato samples was 99.31%, using the back-propagation neural network (BPNN) method (Wan et al., 2018).

Dananjayan et al. used fine-tuned models (e.g., CenterNet, YOLOv4, Faster-RCNN, DetectoRS, and Cascade-RCNN) to detect citrus leaf diseases. Simulation results of their work indicated that the fine-tuned models could classify different stages of citrus leaf diseases. The output of deep CNN-based models showed that CenterNet2 with Res2Net 101 DCN-BiFPN could detect the early stages of citrus leaf diseases (Dananjayan et al., 2022). Syed-Ab-Rahman et al. presented a model that used leaf images for citrus disease detection based on a deep classifier. They employed a region proposal network to propose the potential target infected pixels of the image in order to classify the corresponding disease class. This model achieved 94.37% accuracy in the classification of citrus disease that are citrus black spot, citrus bacterial canker, and Huanglongbing (Syed-Ab-Rahman et al., 2022). Sharif

et al. used a hybrid approach for the prediction of diseases in citrus plants. They employed an optimized weighted image segmentation approach to make enhanced input data. They also applied color, texture, and features selected by the hybrid feature selection approach in order to citrus disease detection. Their technique reached 97% accuracy in detection of citrus disease that are black spot, anthracnose, greening, canker, melanose, and scab (Sharif et al., 2018).

Since deep learning (DL) models depend on data and developing a high-quality training dataset is very challenging and requires a huge amount of data, several data augmentation techniques were utilized to increase the size of datasets. Furthermore, the stability of fine-tuned DL models decreases when the training set has less representative data. However, using fundamental data augmentation techniques for model training, state-of-the-art models might fit too closely to the training dataset, memorizing the noise, and become overfitted. These models do not perform well on new datasets with unseen instances and real-world issues. This study's main goal is to provide a robust and generalized CNN and prevent overfitting of the DL model. Some of the most important objectives of this study are:

- 1) Designing a stabilized model for predicting and accurately diagnosing different ripeness levels (unripe, half-ripe, and ripe) and detecting citrus black spot disease in oranges.
- 2) Increasing robustness of networks by providing a model to generate new noisy images via selecting the optimal parameters of Gaussian, speckle, Poisson, and salt-and-pepper noise.
- 3) Preventing the overfitting problem by applying the best policies of learning-to-augment strategy to fine-tune the pre-trained models with the restored images produced by a convolutional autoencoder.

2. Proposed method

2.1. Dataset

A dataset of 1896 orange images was collected on a citrus farm in Sari, Mazandaran Province, Iran. In the dataset, there were 241 images of unripe oranges, 350 images of half-ripe oranges, 288 images of ripe oranges, and 200 images of oranges infected with black spot disease. Examples of orange images are shown in Fig. 1 in four classes of unripe, half-ripe, ripe, and fruit infected with black spot disease, which experts have classified. A Samsung SM-J500H smartphone's camera (Samsung Electronics, South Korean) was used to capture the orange fruit images.

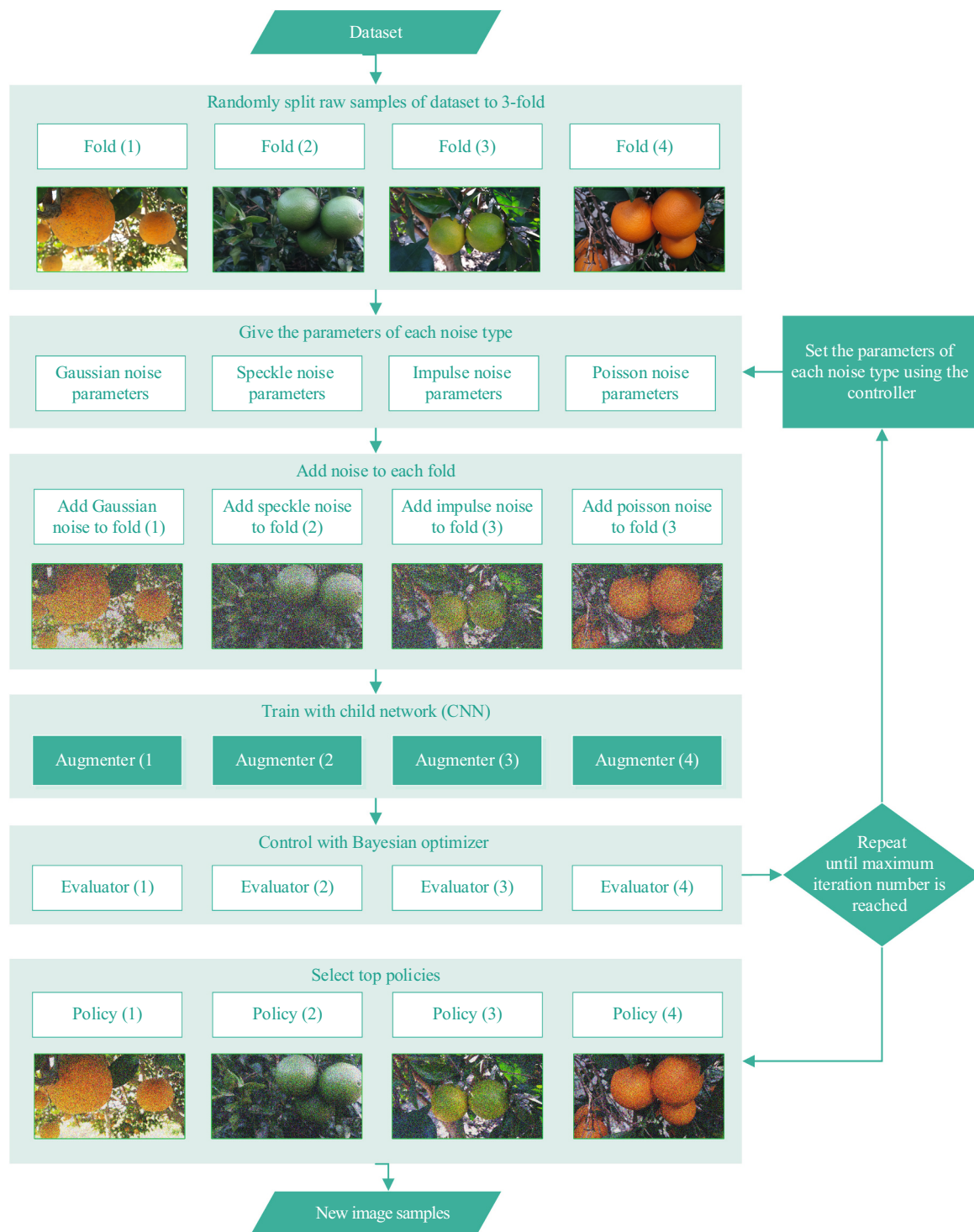


Fig. 3. Flowchart of the proposed noisy-image-based data augmentation model based on the learning-to-augment strategy.

The camera's specifications are presented in Table 1. The images were saved in JPEG file format and converted to the input size for pre-trained CNN. Images were taken from a distance of 30 ± 5 cm from the fruit.

2.2. Learning-to-augment strategy

2.2.1. Data augmentation with noisy images

For the purpose of improving the robustness and generalizability of the deep classifiers during the training phase, digital images, after being added noise, can be applied as augmented data. In this study, an

approach using a learning-to-augment strategy was employed that uses noisy input images with optimal noise parameters to generate new data (Akbarimajd et al., 2022; Momeny et al., 2021b). So that salt-and-pepper noise (Barbosa et al., 2021; Nooshyar and Momeny, 2013), Gaussian noise (Li and Zheng, 2022), speckle noise (Khan and Altalbe, 2022), and Poisson noise (Wang et al., 2021) have been used to create new data for the input of deep CNNs (Bae et al., 2018). Mean and variance were the parameters of speckle and Gaussian noise that should be optimized using the proposed model (Karimi and Taban, 2021). Also, the noise parameter (e.g. noise density) of salt-and-pepper noise can be explored and

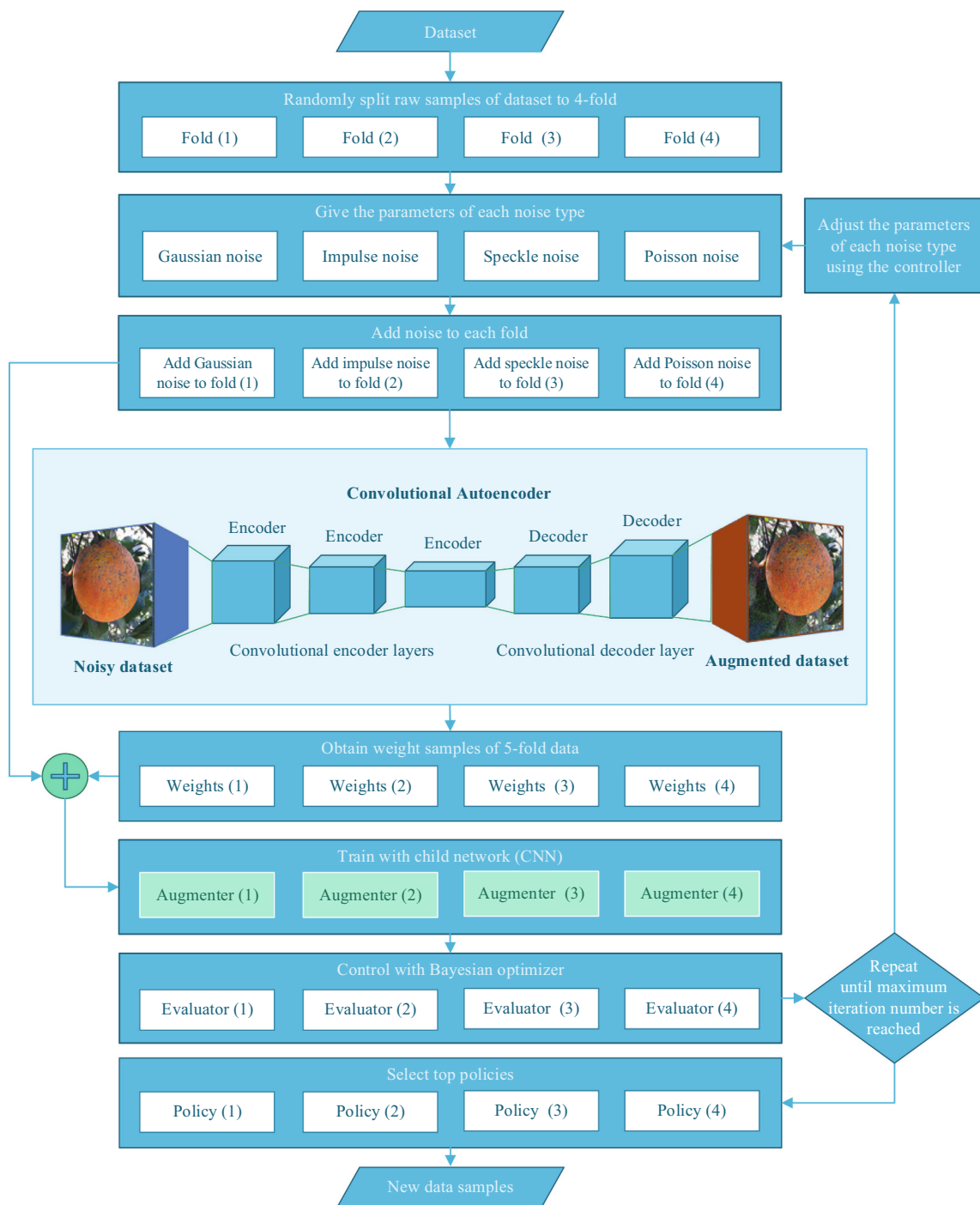


Fig. 4. Schematic workflow of the CAE-based data augmentation model via learning-to-augment strategy.

optimized to produce new augmented data (Wang et al., 2021). The noise type definitions are given in Table 2.

The phases of the noise-based learning-to-augment model are shown in Fig. 2. The schematic workflow of the noisy-image-based data augmentation method is illustrated in Fig. 3. The purpose of this method is to find best policies, optimized parameters of different noise types, to generate new images. As can be seen, first, the input images pool is divided into four folds in order to augment the data. After that, noise is added separately to each fold by a noisy image generator (i.e., salt-and-pepper, speckle, Poisson, and Gaussian noise). To speed up the process of searching for reaching the best policies, we divided the raw samples in the dataset into 4-folds randomly. The child network (i.e., evaluator)

training time for each fold is reduced due to the lower number of samples in the 4-fold splits (Lim et al., 2019). The augmenters create new data based on the new parameters found by the the controller, Bayesian optimization algorithm (Frazier, 2018; Pelikan et al., 1999). Then, the Child CNN (the pre-trained ShuffleNet) model processes each fold. To the results obtained by Child CNNs (the fine-tuned ShuffleNet), the controller optimizes weak policy and keeps firm policy. The controller finds the optimized set of the parameters of each noise type through the Bayesian optimization algorithm. The formula of the Bayesian optimization algorithm can be written as the following (Eq. (1)) if we take $\mathcal{S}$ as the search space and f as the loss function of a child network (Momeny et al., 2021b):

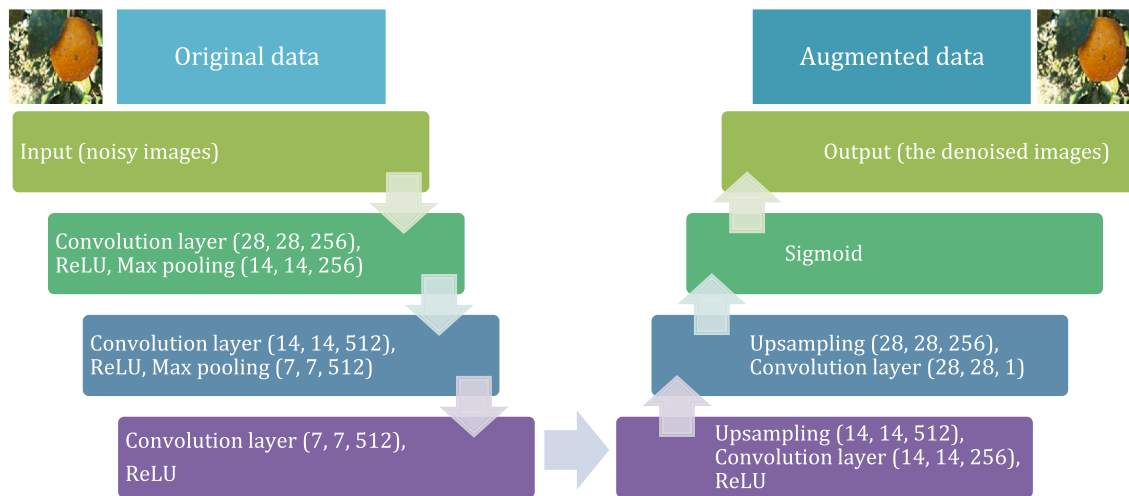


Fig. 5. The proposed CAE model.

Table 3

Properties of the pre-trained models.

Networks	Depth	Size (MB)	Parameters (Millions)
ResNet18	18	44	11.7
GoogleNet	22	27	7.0
ShuffleNet	50	5.4	1.4
ResNet50	50	96	25.6
MobileNetv2	53	13	3.5
DenseNet201	201	77	20.0

$$y = \arg \min_G(G), G \in \mathcal{G} \quad (1)$$

The aim of the optimization problem in Eq. (1) is to find y to minimize $f(G)$ for G in a bounded domain $\mathcal{G}$. The loss function for the Bayesian optimization algorithm is calculated through the losses of the child CNNs (the fine-tuned ShuffleNet). The process of noisy-image-based data augmentation repeats until finding the maximum iteration number.

2.2.2. Data augmentation approach with restored data

The proposed model uses an automatic augmentation policy search

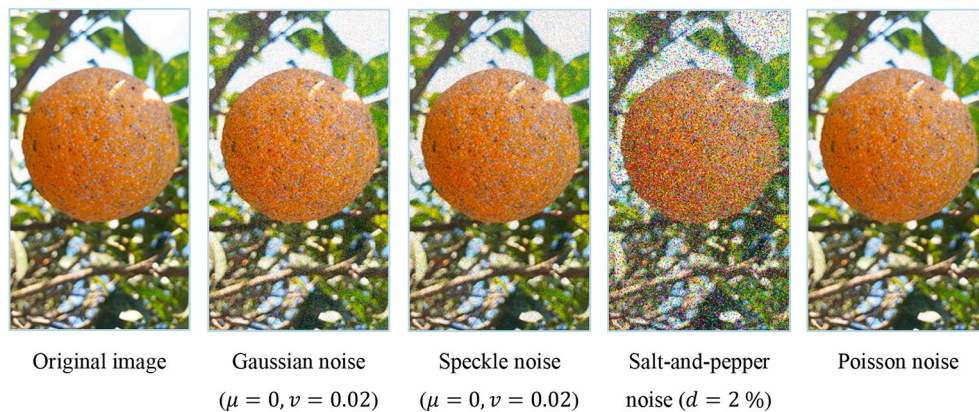


Fig. 6. The noise was added to the orange fruit images.

	Infected	Unripe	Half-ripe	Ripe
Step 1	Positive	Negative	Negative	Negative
Step 2	Negative	Positive	Negative	Negative
Step 3	Negative	Negative	Positive	Negative
Step 4	Negative	Negative	Negative	Positive

Fig. 7. Performance evaluation of the detection of citrus black spot disease and fruit ripeness level by assigning the positive and negative classes in four steps.

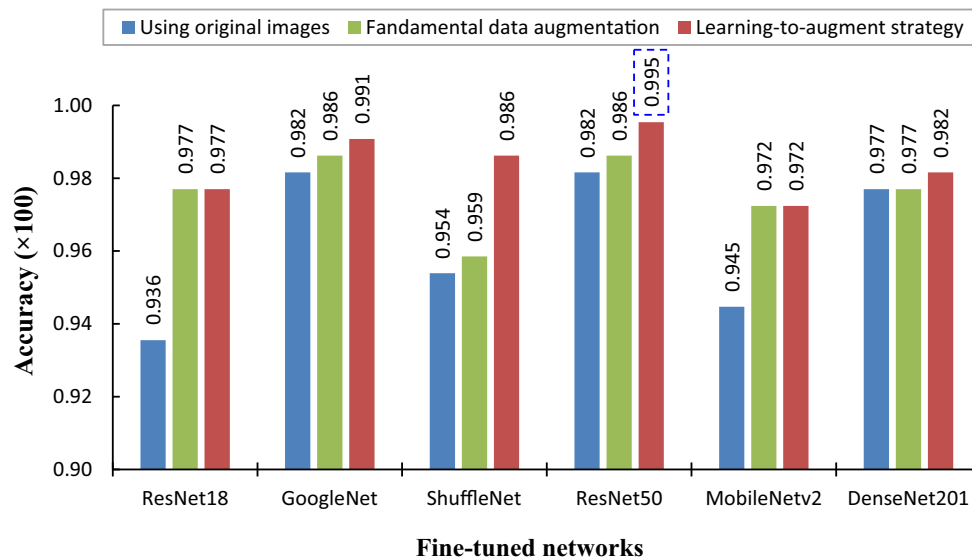


Fig. 8. The accuracies of detection of citrus black spot disease and fruit ripeness level for different fine-tune approaches.

approach by restoring images that were purposefully affected by salt-and-pepper, Gaussian, speckle, and Poisson noise. It is possible to restore noisy images through image enhancement algorithms such as a convolutional autoencoder (CAE) (Zhang et al., 2020). However, the value of pixels in the restored (denoised) image and the original one are not equal due to noise types and densities (Guo et al., 2020). Employing the learning-to-augment method using restored data, we apply the dissimilarity between the restored pixels and the original pixels to generate new data. Initially, we add specific noise and parameters to the dataset's images. After that, by using the proposed CAE model, we partially remove the noise from the image. The CAE aims to produce the augmented data from the noisy image input, where the original images are set as the target. In the end, the restored images are used as new augmented data (Momeny et al., 2021b).

The type and parameters of noise are important factors in using denoising images as a data augmentation model. The restored image can be very similar to the original one depending on the error rate of the noise removal and image enhancement methods, especially in the case of low-density noise, and make an ineffective augmented image. However, high-density noise with a weak denoising level causes the value of pixels in the restored images to be more dissimilar to the original images (Momeny et al., 2021b). With an overview of the process in Fig. 4, the noise generator adds the noise to the original images with given parameters (from the controller). Then, the proposed CAE-based denoising model takes noisy samples as inputs.

When the CAE is trained, new augmented images are produced from the noisy input data. In this case, again, the new augmented data is fed as input to the child CNN (pre-trained ShuffleNet) models, like in the noisy image-based data augmentation method (Fig. 3). Optimal noise parameters are found through the Bayesian optimization algorithm. Once the optimal policies are found, the whole dataset and the proposed CAE model are applied to augment the data (Fig. 4).

2.3. Computational platform and implementation details

MATLAB and the deep learning toolbox 2022b (MathWorks, Inc., Natick, Massachusetts, United States) to train and fine-tune the proposed models. The provided dataset was divided into three individual sets: 70% for 'training', 10% for 'validation', and 20% for 'testing'. The stochastic gradient descent (SGD) with the learning rate of $1e-5$, minimum batch size of 10, and maximum epoch of 10 was used to fine-tune the pre-trained networks. The Adam optimizer with a learning rate of

0.001 was used to train the CAE, and the mean squared error (MSE) was applied as the loss function. The configuration of CAE can be seen in Fig. 5.

For the detection of citrus black spot disease and fruit ripeness level, pre-trained CNNs were used and fine-tuned on the orange image dataset. The weights of layers have not been frozen during fine-tuning process of the pre-trained networks. The performance of six fine-tuned models, ShuffleNet (Zhang et al., 2018), ResNet18 (He et al., 2016), ResNet50 (He et al., 2016), MobileNetv2 (Sandler et al., 2018), DenseNet201 (Huang et al., 2017), and GoogleNet (Szegedy et al., 2015) was then evaluated. The depth, size, and parameters of networks used in the proposed framework are summarized in Table 3.

Fig. 6 illustrates the output of the noisy image generators (Fig. 3), which separately added Gaussian, speckle, salt-and-pepper, and Poisson noise to the images of unripe, half-ripe, ripe, and infected with black spot disease orange image cases in each fold.

In order to fine-tune the pre-trained networks, the data augmentation policies obtained with the learning-to-augment method using noisy and restored images were processed. The fundamental data augmentation approaches, image rotation of 180 degrees and flipping the image along the vertical axes were used for comparison. To assess the results of the fine-tuned deep networks, we applied six metrics for verifying the quality of the detection of citrus black spot disease and classification of fruit ripeness level, including accuracy, sensitivity, specificity, precision, recall, and F-measure criteria. As seen in Fig. 7, there were four steps to evaluate the classification task in terms of the mentioned metrics. At each assessment and evaluation phase, one class was set as positive, and the others were set as negative. For example, in step 1, the images infected with black spot disease were considered as the positive class and the other three (unripe, half-ripe, and ripe images) as negative.

3. Results

The comparison of classification for the fine-tuned networks with and without the learning-to-augment strategy has been illustrated in Fig. 8. The results indicate that the ResNet50 fine-tuned with the proposed model achieved the best accuracy (99.5%). Therefore, the present results indicate that the proposed ResNet50 method with learning-to-augment strategies as a simple, fast, and non-destructive method can be useful in disease management (diagnosis of black spot disease) and detection of ripening level of orange fruit. Training accuracy and loss curves of a pre-trained network with the parameters in the detection of

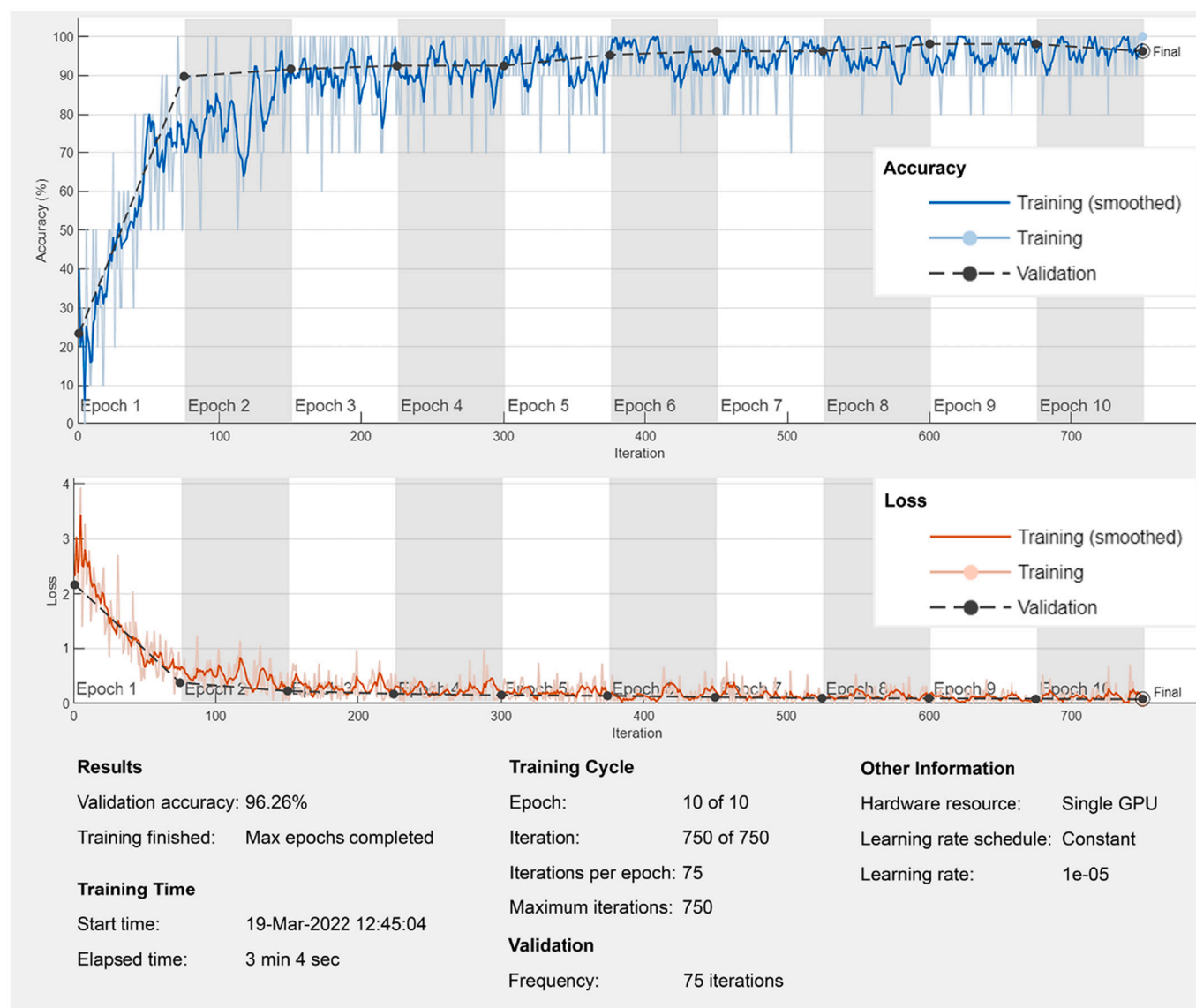


Fig. 9. Training accuracy and loss curves of ShuffleNet with network parameters in the detection of citrus black spot disease and fruit ripeness level.

citrus black spot disease and fruit ripeness level are illustrated in Fig. 9.

The confusion matrix related to the grading and diagnosis of black spot disease and the ripeness level of orange fruit is reported using the proposed learning-to-augment model in Fig. 10. The results of the confusion matrix show that the ResNet50 model performed better than the other models; only one instance of the half-ripe class was incorrectly classified in the unripe class. The classification accuracy of the ResNet50 model was 100% for the black spot disease, half-ripe, and ripe classes, and 98% for the unripe class. Therefore, the results of this study indicate that the proposed learning-to-augment method and the developed predictive models can be useful tools for managing and controlling black spot disease and classifying fruit ripeness levels. These results are similar to the studies reported by Khojastehnazhand et al. and Mutha et al. for grading apricot and tomato fruit via maturity detection (Khojastehnazhand et al., 2019; Mutha et al., 2021).

The test results in terms of sensitivity, specificity, precision, recall, and F-measure metrics are presented in Table 4. As seen in Table 4, the detection of citrus black spot disease and ripeness level in orange fruit by the proposed approach (applying the learning-to-augment method) by the ResNet50 method are better than other pre-trained networks augmented by fundamental data augmentation approaches. For

example, considering images of oranges infected with black spot disease as the positive class and the others (unripe, half-ripe, and ripe images) as negative, the higher performance of the proposed classification methodology can be seen in Table 4 via the evaluation metrics. The 100% sensitivity value in the positive class indicates that all infected images were classified correctly. The specificity value shows that no samples from non-infected classes were misclassified. The same is true when ripe images have been considered as a positive class.

Because of existing differences in the distribution of the classes in the provided dataset (241 images of unripe, 350 of half-ripe, 288 of ripe, and 200 of infected), the F-measure was used to compare the results of the models. Using confusion matrices for detecting citrus black spot disease and fruit ripeness level by the proposed learning-to-augment model, the F-measure was calculated as the harmonic mean of precision and recall, giving each the same weighting. It allows a model to be evaluated, taking both the precision and recall into account using a single score, which is helpful when describing the proposed model's performance and comparing models. Looking at the maximum F-measure for the fine-tuned networks with learning to augment strategy shows that ResNet50 performed significantly better than the other methods (Fig. 11).

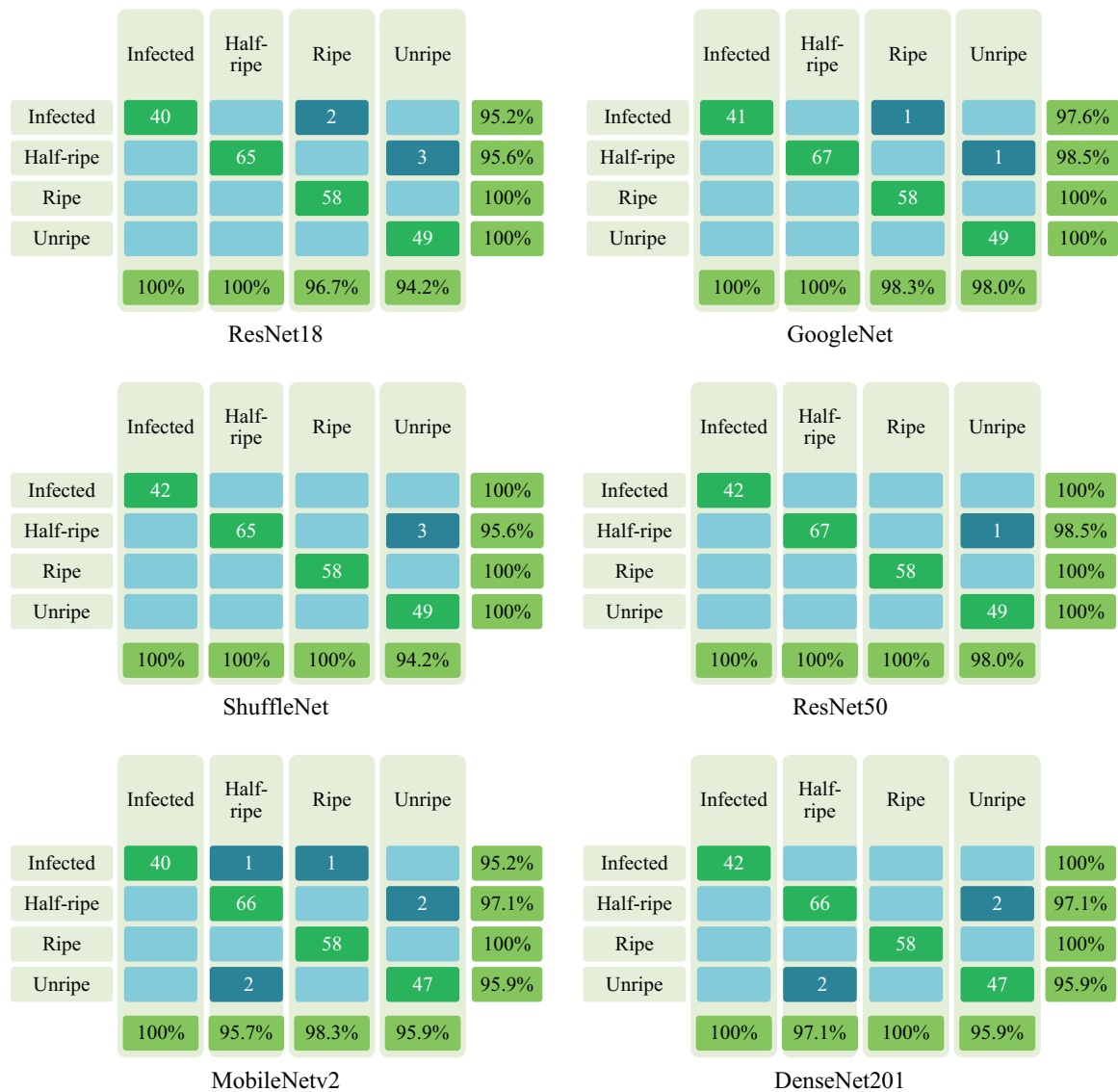


Fig. 10. Confusion matrices for detecting citrus black spot disease and fruit ripeness level by the proposed learning-to-augment model.

4. Discussion

Fruit (like oranges) have many nutritional benefits and usually contain several essential vitamins and minerals for humans that are not found in other foods. Eating healthy fruit energizes humans and delivers a wide range of vitamins, minerals, and antioxidants to the human body, ultimately improving their health. Therefore, due to the importance of fruit consumption, achieving simple, rapid, and non-destructive methods for quality control and fruit grading is of particular importance. Different qualities of an orange fruit (such as unripe, half-ripe, ripe, and infected oranges) must be identified and classified before packing. Due to the unique characteristics of orange fruit and many operations such as harvesting, warehousing, transportation, etc. carried out to deliver the product from the field to be consumed, providing a healthy and high-quality product is challenging. Therefore, according to the quality of the fruit in terms of ripeness and disease-free, orange fruit can be graded from the tree to reduce waste and increase marketability, and then harvest only the healthy, high-quality fruit. This study showed that image processing techniques and the CNN method were very effective and useful for this type of research. The methods used in this research can identify citrus black spot infected fruit, and then the farmer

can prevent their losses by performing targeted spraying operations. Development of fruit disease detection and classification system can be done in perspective of remote monitoring of the crop to take timely decision and improve fruit quality. Real-time and reliable pest classification and detection are very important in pest and disease monitoring for fruit protection (Abdulridha et al., 2019). Based on the recognition of orchard pests and fruit quality, this study can further improve the efficiency of precision agriculture practices, such as optimizing/reducing the use of pesticides and waste control, so as to play a role in ecological protection and the maintenance of ecological balance. Deep learning, especially the development of CNN models, have brought many breakthroughs in the field of intelligent information processing in pest and disease monitoring and reveals great potential for complementing traditional laborious inspection (Li et al., 2021; Naik et al., 2022; She et al., 2022; Wang et al., 2022). The proposed automatic disease and fruit quality monitoring system can be used not only in agriculture but also be extended to the automatic monitoring and control of forest pests and diseases.

In similar research, Shin et al. for the diagnosis of healthy and infected leaves of strawberries using the ResNet-50 model, achieved an accuracy of 98.11% (Shin et al., 2021). Ashtiani et al. achieved a 98.03%

Table 4
Comparing sensitivity, specificity, precision, recall, and F-measure for detection of black spot disease and ripeness level of citrus fruit.

Method	Positive class	Using original data without data augmentation (%)					Applying fundamental data augmentation (%)					Applying learning-to-augment strategy (%)				
		Sensitivity	Specificity	Precision	Recall	F-Measure	Sensitivity	Specificity	Precision	Recall	F-Measure	Sensitivity	Specificity	Precision	Recall	F-Measure
ResNet18	Infected	90.5	98.9	95.0	90.5	92.7	97.6	100	100	97.6	98.8	95.2	100	100	95.2	97.6
	Unripe	93.9	95.8	86.8	93.9	90.2	98.0	98.2	94.1	98.0	96.0	100	98.2	94.2	100	97.0
	Half-ripe	92.6	98.7	96.9	92.6	94.7	100	99.4	98.3	100	99.1	95.6	100	95.6	95.6	97.7
GoogleNet	Ripe	96.6	98.1	94.9	96.6	95.7	95.6	99.3	98.5	95.6	97.0	100	98.7	96.7	100	98.3
	Infected	95.2	99.4	97.6	95.2	96.4	97.6	100	100	97.6	98.8	97.6	100	98.0	97.6	98.8
	Unripe	100	99.4	98.0	100	99.0	100	98.8	96.1	100	98.0	100	99.4	98.0	100	99.0
ShuffleNet	Half-ripe	98.5	100	100	98.5	99.3	100	99.4	98.3	100	99.1	98.5	100	98.3	100	99.3
	Ripe	98.3	98.7	96.6	98.3	97.4	97.1	100	100	97.1	98.5	100	99.4	98.3	100	99.1
	Infected	92.9	100	100	92.9	96.3	95.2	100	100	95.2	97.6	100	100	100	100	100
ResNet50	Unripe	91.8	100	100	91.8	95.7	95.9	97.0	90.4	95.9	93.1	100	98.2	94.2	100	97.0
	Half-ripe	97.1	96.0	91.7	97.1	94.3	100	98.7	96.7	100	98.3	95.6	100	100	95.6	97.7
	Ripe	98.3	97.5	93.4	98.3	95.8	92.6	98.7	96.9	92.6	94.7	100	100	100	100	100
MobileNetV2	Infected	97.6	100	100	97.6	98.8	97.6	100	100	97.6	98.8	100	100	100	100	100
	Unripe	100	98.2	94.2	100	97.0	100	98.8	96.1	100	98.0	100	99.4	98.0	100	99.0
	Half-ripe	95.6	100	100	95.6	97.7	100	99.4	98.3	100	99.1	98.5	100	100	98.5	99.3
DenseNet201	Ripe	100	99.4	98.3	100	99.1	97.1	100	100	97.1	98.5	100	100	100	100	100
	Infected	95.2	99.4	97.6	95.2	96.4	97.6	100	100	97.6	98.8	95.2	100	100	95.2	97.6
	Unripe	93.9	97.0	90.2	93.9	92.0	95.9	98.2	94.0	95.9	94.9	95.9	98.8	95.9	95.9	95.9
DenseNet201	Half-ripe	92.6	97.3	94.0	92.6	93.3	100	99.4	98.3	100	99.1	97.1	98.0	95.7	97.1	96.4
	Ripe	96.6	98.7	96.6	96.6	96.6	95.6	98.7	97.0	95.6	96.3	100	99.4	98.3	100	99.1
	Infected	97.6	100	100	97.6	98.8	100	100	100	97.6	98.8	97.6	100	98.3	97.6	98.8
DenseNet201	Unripe	95.9	98.8	95.9	95.9	95.9	95.9	98.2	94.0	95.9	94.9	95.9	98.8	95.9	95.9	95.9
	Half-ripe	97.1	98.7	97.1	97.1	97.1	100	100	100	100	100	97.1	98.7	97.1	97.1	97.1
	Ripe	100	99.4	98.3	100	99.1	95.6	98.7	97.0	95.6	96.3	100	99.4	98.3	100	99.1

accuracy in mulberry ripeness stages classification using the ResNet-18 model too (Ashtiani et al., 2021). Huang et al. in a similar study, reported that the ResNet50 model was able to detect tomato pests with 97.12% accuracy (Huang et al., 2022). Yu et al. in diagnosing tomato pests, reported that among six networks evaluated in an experiment, ResNet50 has an average diagnostic accuracy of 94.4% for seven tomato pests, which was better than other models (Yu et al., 2022).

In a study to control waste, Jahanbakhshi et al. utilized a generalized CNN by configuring the Batch Normalization (BN)-CNN method based on mixed pooling to grade carrots and reported 99.43% accuracy (Jahanbakhshi et al., 2021c). Momeny et al. classified cherry fruit using a method capable of improving generalization in the CNN by combining max pooling and average pooling. To avoid overfitting, they used data augmentation to generate additional data and increase the CNN accuracy. They reported that the proposed model was able to classify cherry fruit with 99.4% accuracy (Momeny et al., 2020). Jahanbakhshi et al. used a stochastic pooling mechanism and data augmentation to improve CNN, prevent overfitting, and then classify sour lemon fruit with 100% accuracy (Jahanbakhshi et al., 2020). Kausar et al. used Global Average Pooling (GAP) to classify fruit images to minimize the number of parameters and protect the model from overfitting. Their results showed that for 55,244 color fruit images from 81 categories, the Pure-CNN achieved 98.88% classification accuracy (Kausar et al., 2018).

The images used in the present research were taken under uncontrolled conditions regarding lighting conditions and camera stabilization. The proposed robust and generalized model described in our study provides accurate assistance in identifying citrus black spots and levels of fruit ripening, which could help waste control and reduce economic and crop yield losses. Applying noisy and restored images as augmented data using the parameters optimized by the learning-to-augment strategy can reduce the overfitting problem and improve the robustness and generalizability of fine-tuned networks. As a result, the proposed method in this study exhibits a high accuracy of 99.5% (Fig. 9).

5. Conclusion

In this paper, we proposed a stabilized, fine-tuned model via the learning-to-augment strategy using orange fruit images affected by noise and restored with an autoencoder for the detection of citrus black spot disease and ripeness level of orange fruit and classifying them into four classes (unripe, half-ripe, ripe, and infected with black spot disease). Using an effective noise-based data augmentation approach (i.e., Gaussian, speckle, Poisson, and salt-and-pepper noise), a more robust CNN model was developed instead of the fundamental data augmentation method to avoid overfitting. The proposed model stabilized the learning process of deep classifiers by employing a noisy image generator based on optimized noise parameters, a controller using a Bayesian optimization algorithm, a convolutional autoencoder network with a noisy image input, and child networks (pre-trained ShuffleNet). The fine-tuned ResNet50 model utilizing learning-to-augment using noisy and restored data showed the best performance (100%) in terms of sensitivity, specificity, precision, recall, and F-measure when classifying the fruit infected with black spot disease. It can be concluded from the results that the learning-to-augment strategy improves the robustness and generalization of the deep fine-tuned networks.

Declaration of Competing Interest

None.

Data availability

The authors do not have permission to share data.

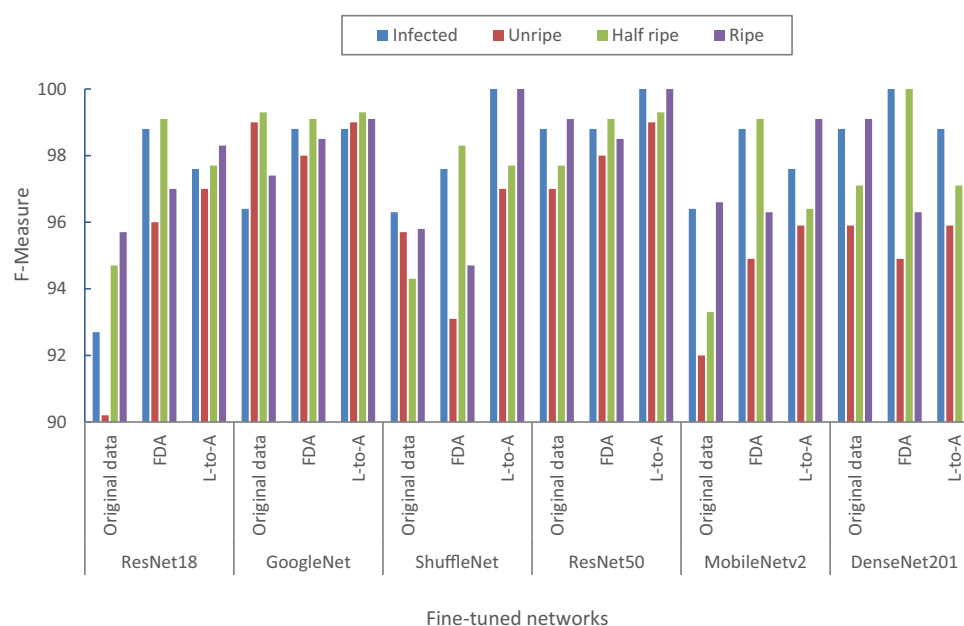


Fig. 11. Comparing the detection of citrus black spot disease and fruit ripeness level for the pre-trained networks that were fine-tuned with different approaches. The images, including fruit infected with black spot disease, unripe, half-ripe, and ripe fruit, are considered the positive classes in four steps. Here, 'FDA' represents the 'Fundamental Data Augmentation Techniques' and 'L-to-A' represents the 'Learning-to-Augment Strategy'.

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